Tosco Branch Re-Entry Wells

Production Potential Based on Execution

Bowman County, North Dakota • Williston Basin southern margin (Red River trend)

BRANCH 4 RR

NDIC #41700 • API 33-011-01562

Status: DRILLING (06/19/2026)

Re-entry • short-radius (25-30 deg/100ft)

Sec 19-129N-103W

BRANCH 2

NDIC #41258 • permitted 06/11/2026

Status: PERMITTED (re-entry)

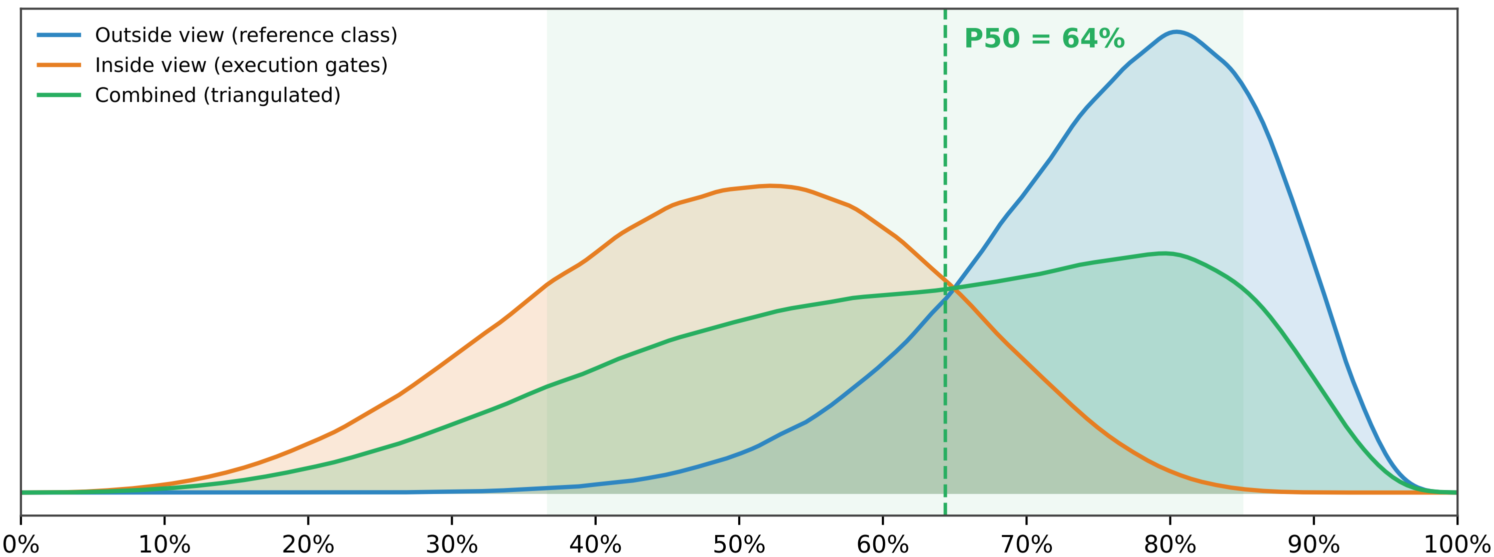
Re-entry into same Marlo pool

BHL Sec 20-129N-104W

Bottom line

Both wells are RE-ENTRIES into a pool that Marlo just discovered and is already producing (sister well Branch 5, NDIC #41701). That reclassifies them from 'wildcat' (~20% success) to DEVELOPMENT-CLASS step-outs. The reservoir is effectively de-risked; the bet is EXECUTION of the short-radius re-entry. Same intrinsic potential for both wells — Branch 4 is further along (drilling now), Branch 2 is at the start (permitted), so Branch 4's risk is retiring first.

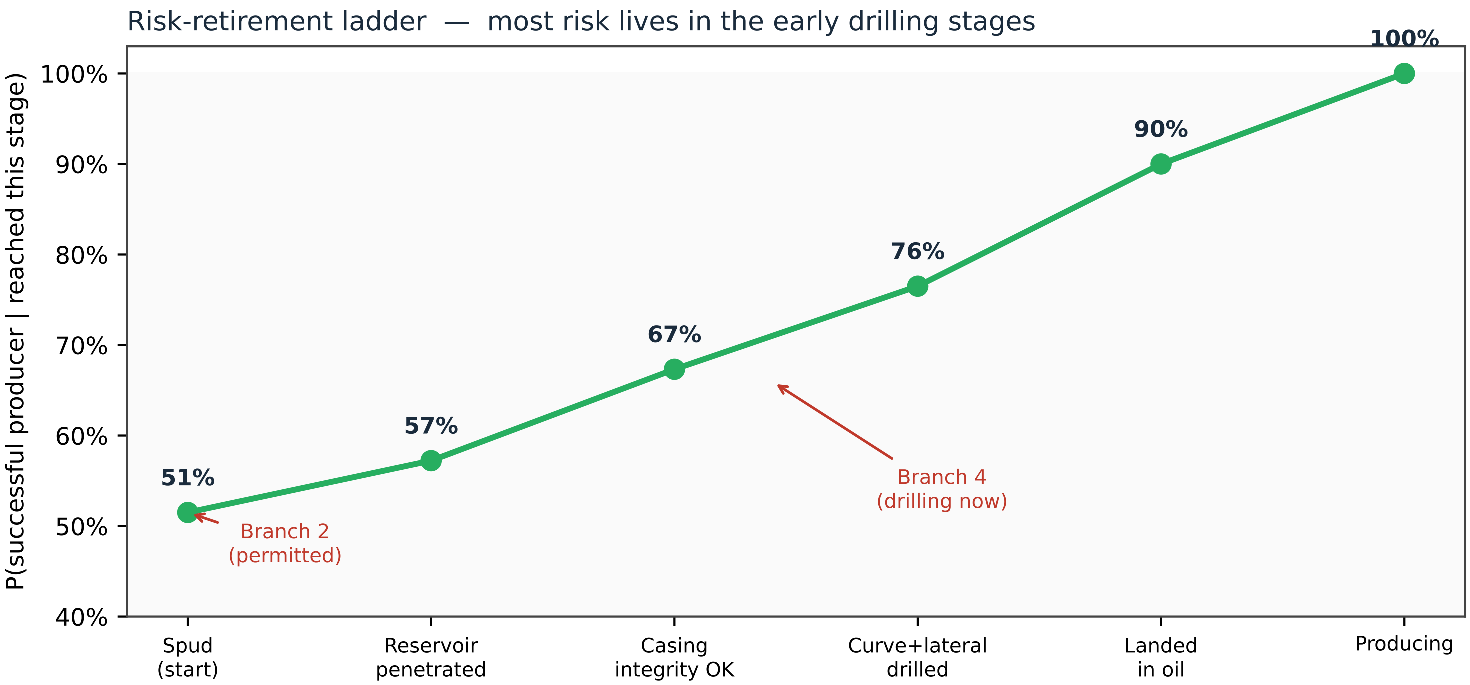
Chance of success — three independent estimates



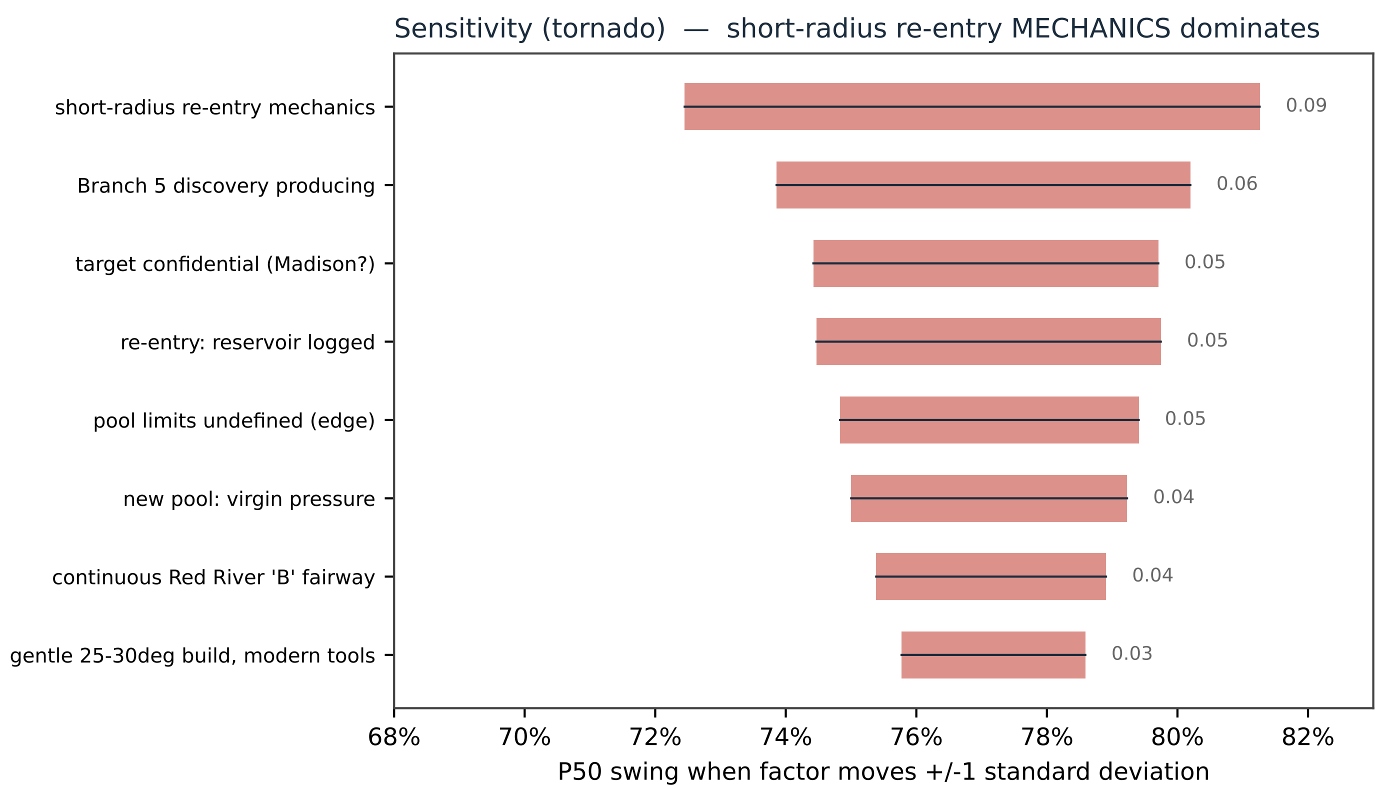
Probability of a successful producing well (combined P50 64%; 80% band 37%-85%)

Dynamics of Execution — how risk retires

Probability climbs as each well passes its drilling and completion gates.



Read: the biggest lever is mechanical execution of the re-entry — the risk the original wellbore's casing/cement-bond log can resolve before the curve is ever drilled.



The Mathematics, Explained

A transparent two-estimator Bayesian Monte-Carlo. Inputs are structured judgment in log-odds; the reference class is empirical.

$$p_0 \sim \mathrm{Beta}(s+1,\, f+1)$$

Reference-class prior. Start from how often analog wells actually succeeded: s successes, f failures (NDGS Red River development/extension = 20 of 27). The empirical anchor.

$$p = \sigma\Big(\mathrm{logit}(p_0) + \sum_i \Delta_i\Big), \quad \sigma(x) = \frac{1}{1+e^{-x}}$$

Outside view. Combine evidence in LOG-ODDS (logit): each factor delta_i nudges the odds up (re-entry, adjacent producer) or down (re-entry mechanics, confidential target). sigma maps back to a probability.

$$P_{\mathrm{in}} = \prod_{k=1}^{5} g_k, \quad g_k = \sigma\big(\mathrm{logit}(b_k) + \lambda_k^e q_e + \lambda_k^g q_g + \varepsilon_k\big)$$

Inside view. A chain of 5 execution gates (reservoir -> casing -> curve -> placement -> rate). Two SHARED latent factors q_e (execution quality) and q_g (geology) make the gates rise and fall together (correlation).

$$\mathbb{E}[\sigma(\mathrm{logit}(b) + \varepsilon)] < b \quad (b > \tfrac{1}{2})$$

Jensen's inequality. The logistic is concave above 0.5, so UNCERTAINTY drags each gate's expected pass-rate BELOW its nominal value. This is why the simulated inside-view mean (0.49) sits just under the naive product (0.515).

$$\{p\} = \{p_{\mathrm{out}}\} \cup \{p_{\mathrm{in}}\} \;\Rightarrow\; P_{10},\, P_{50},\, P_{90}$$

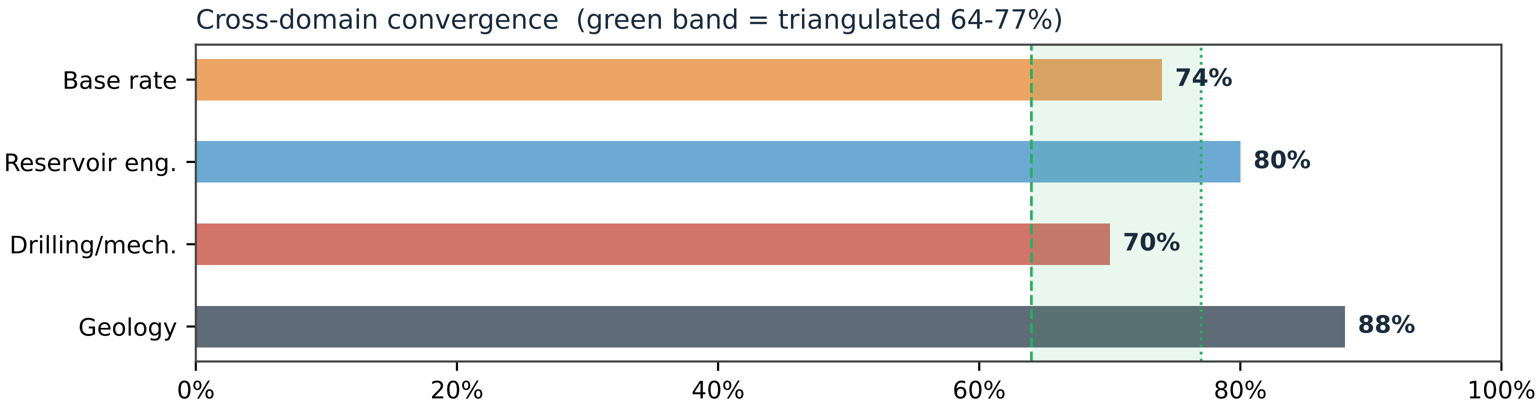
Triangulation. Pool the two independent sample sets and report the distribution. Their divergence (0.77 vs 0.50) is diagnostic: it localizes the uncertainty to execution.

$$P(\mathrm{success} \mid \mathrm{gate}\ k\ \mathrm{passed}) = \prod_{j>k} b_j$$

Dynamic update (Bayes). Once a gate is observed to pass, drop its risk; the conditional probability climbs (the ladder on p.2: 51% -> 57% -> 67% -> 77% -> 90%).

Verification, Validation & Sources

Four independent domains converge; verified (no leakage) and validated against field analogs.



Verified: all model numbers reconcile with the Monte-Carlo output; forward-generative (no fitted predictor, no data leakage); inside/outside divergence reported, not averaged away. Independently re-checked by a fresh-context verifier (one propagated wording error on the Jensen effect was found and corrected). Validated: the DOE/Luff Bowman re-entry program (0 of 4 dry from no-reservoir, ~1 of 4 clearly economic) corroborates BOTH the high reservoir CoS and the heavy mechanical/placement discount.

Selected citations

[1] NDIC Oil & Gas Division — Daily Activity Report 06/19/2026 (File #41700, re-entry, DRILLING); Hearing Docket 04/22/2026, Case #32729 (Branch 5 discovery, field-limit spacing). dmr.nd.gov/oilgas

[2] N. Dakota Geological Survey, 'Oil & Gas Potential of the Red River Fm, SW North Dakota' (2017) — development/extension success 70-86% (Camel Hump 6/7; step-out 14/20).

[3] Carrell, George & Gibbons (1997), DOE / Luff Exploration, 'Lateral Drilling & Completion Tech., Red River & Ratcliffe, Williston Basin', OSTI #2214 — short-radius re-entry case results & costs.

[4] Oil & Gas Journal, 'Horizontal projects buoy Williston recovery' — Cedar Hills Red River 'B' (IP, EUR, RF).

[5] AAPG Datapages — 'The Red River B Zone at Cedar Hills Field, Bowman County, ND' (reservoir character).

[6] USGS — 'Geology & Undiscovered Oil & Gas Resources, Madison Group, Williston Basin' (Mission Canyon).

[7] DrillingEdge / ShaleXP — Bowman County & Marlo Operating activity (Branch 5 producing, Apr 2026).

[8] Reference-class forecasting / 'outside view' — Kahneman & Tversky (1979); Flyvbjerg (2006).

Scope: probability of a producing well, NOT economics. Inputs are expert-judgment priors except the empirical reference class. Target formation confidential to 12/19/2026; assessment assumes the Red River trend. Magnitude if successful: IP ~tens-to-low-hundreds bopd (offset Branch 5 ≈ 65 bopd), EUR ~40–150 Mbbl.